MARTIN DESSART

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Research and Engineering Experience

10/2024 – 08/2026
Department of Biochemistry
Virginia Tech - Blacksburg, VA

Postdoctoral Research in Neuroethology
Investigated sensory integration and chronobiology in mosquitoes.
Mentor: Dr. Clément Vinauger

02/2021 – 02/2024
IRBI – UMR 7261, CNRS
University of Tours
Tours, France

Research in Neuroethology and Sensory Ecology
Examined the impact of pollutants on mosquito larvae cognition.
Assessed the role of larvae as bioindicators of freshwater ecosystems.
Supervisors: Prof. Claudio Lazzari & Dr. Fernando Guerrieri

07/2020 – 10/2020
CRCA, GEODE, CNRS
University Paul Sabatier
Toulouse, France

Master's Thesis Internship in Neuroethology
Collected flower reflectance in the field using spectrophotometry.
Modeled flower visual saliency using honeybee-vision models.
Supervisors: Dr. Sylvie Guillerme & Prof. Martin Giurfa

01/2020 – 07/2020
Animal Minds Lab
University of Auckland
Christchurch, New Zealand

Master's Thesis Internship in Neuroethology
Designed and conducted judgment bias tests on kea *Nestor notabilis*.
Analyzed keas vocalizations using Raven and Praat softwares.
Supervisors: Dr. Alexander Taylor & Dr. Amalia Bastos

04/2019 – 06/2019
UMR MARBEC, CNRS
University of Montpellier
Montpellier, France

Research Internship in Molecular Biology
Designed an experiment on fish gut methane emissions.
Performed field sampling and molecular analyses.
Supervisors: Dr. Jean-Christophe Auguet & Dr. M.C. Cheutin

09/2016 – 02/2017
Auckland University
Auckland, New Zealand

Engineering Thesis Internship in Computing
Analyzed wind-farm electricity generation.
Supervisors: Prof. Tek Tjing Lie & Dr. Gianni Goretti

05/2015 – 08/2015
TESEO – Clemessy Entity
Torino, Italy

Engineering Internship in Computing
Automated a maintenance procedure using VBA macros.
Conducted radar calibration in anechoic chamber.
Supervisor: Jean-Louis Menu

09/2014 – 09/2017
Clemessy Aeronautics & Space
Toulouse, France

Engineering Apprenticeship in Control Command
Designed and integrated automated cable-testing system.
Supervisor: Jean-Paul Herrgott

Education

02/2021 – 02/2024
University of Tours
Tours, France

PhD at Insect Biology Research Institute
Designed and conducted experiments in cognition, ecotoxicology.
Performed data analysis, programming, field work, teaching.

09/2018 – 09/2020
University of Strasbourg
Strasbourg, France

2 Years Master Program (High Honours)
Studied neurobiology, ecophysiology, ethology, cognition.
Developed skills in statistical analyses and data treatment.

11/2017 – 08/2018
Udacity & Google School, online

Front-End Web Developer Nanodegree
Developed front-end designs and learned how to use frameworks.

09/2014 – 09/2017
Institut Catholique des Arts et Métiers
Toulouse, France

Engineer's Degree – Automation & Control Systems
Studied programming, project management, electronics engineering.

09/2012 – 09/2014
Institut Catholique des Arts et Métiers
Toulouse, France

Preparatory program to the Grandes Écoles institution
Studied electrical engineering, automation, industrial organization.

Teaching and Mentoring

Mentoring

10/2024 – 08/2026
Virginia Tech

Srija Aluri — 05/26 - 08/26 (Sophomore, **Biochemistry**)
Sami Sheik — 01/26 - 06/26 (Junior, **Neuroscience**)
Maya Gosai — 01/26 - 06/26 (Sophomore, **Neuroscience**)
Ashwin Ajai — 01/26 - 08/26 (Sophomore, **Computer Science**)
Jasmine Adams — 09/25 - 08/26 (Senior, **Neuroscience**)
Hayden Sunman — 01/25 - 06/25 (Freshman, **Biochemistry**)
Lily Smith — 01/25 - 06/25 (Senior, **Neuroscience**)
Sydney Luff — 11/24 - 08/26 (Junior, **Biology & Entomology**)
Richard Rust — 11/24 - 05/26 (Graduate, **Neuroscience**)

03/2021 – 05/2021
University of Tours

Tristan Robineau (MSc Student, **Ecology & Evolution**)
Effect of light and turbidity on habituation of *Aedes aegypti* larvae.

06/2017 – 08/2027
Clemessy Aeronautics & Space

Eva Daoud (Engineering Student, **ESTACA Engineering School**)
Database design & conception

Teaching

2026
Virginia Tech

Co-instructor: BCHM 4354 Biochemical Communications
Lecture: Biochemical Communications (15h)
Article discussion and group project supervision (30h)

2021 – 2023
University of Tours

Lecturer
Laboratory session: Arthropod identification (32h)
Lecture: Neuroscience: conditioning and addiction (6h)
Lecture: Arthropod systematics (7.5h)
Laboratory session: Ethology (25h)

Publications

Published

2026 **Dessart M.**, Lazzari C.R., Guerrieri F.J. (2026). *Acute and chronic sublethal chemical pollution affects locomotion, learning and memory in mosquito larvae.* Journal of Experimental Biology; jeb.250493. https://doi.org/10.1242/jeb.250493.

2024 **Dessart M.**, Lazzari C.R., Guerrieri F.J. (2024). *Habituation leads to short but not long-term memory formation in mosquito larvae.* Journal of Insect Physiology 155:104650. https://doi.org/10.1016/j.jinsphys.2024.104650

Dessart M., Aguiar J.M.R., Tabacchi E., Guillerme S., Giurfa M. (2024). *Color-advertising strategies of invasive plants through the bee eye.* Frontiers in Plant Science. 15, 1393204. https://doi.org/10.3389/fpls.2024.1393204

2023 **Dessart M.**, Piñeirúa M., Lazzari C.R., Guerrieri F.J. (2023). *Assessing learning in mosquito larvae using video-tracking.* Journal of Insect Physiology. 149, 104535. https://doi.org/10.1016/j.jinsphys.2023.104535

In preparation

a. **Dessart M.**, Luff S., Smith L., Sunman H., Vinauger C. Rapid phase resetting of *Aedes aegypti* circadian rhythms by transient alterations in light exposure. BioRxiv https://doi.org/10.64898/2026.08.27.747523

b. **Dessart M.**, Rust R., Smith L., Vinauger C. *Visual context modulates threat detection in Aedes aegypti.* In preparation

c. Wynne N., Aronson H., **Dessart M.**, Luff S., Applebach E., Ajayi O., Chandrasegaran K., Benoit J., Lahondère C., Vinauger C. *Larval growing conditions affect the daily rhythms and thermal preferences of adult Aedes japonicus.* In preparation

d. **Dessart M.**, Rust R., Adams J., Wynne N., Chandrasegaran K., Wolff G., Vinauger C. *Modulation of mosquitoes' responses to visual threats.* In preparation

Conference Presentations

2026 **Dessart M.**, Luff S., Smith L., Sunman H., Vinauger C. Quantifying circadian resilience to light disruption in the urban mosquito *Aedes aegypti* – **International Congress of Neuroethology**, Vancouver, Canada — ***Oral presentation***

2025 **Dessart M.**, Rust R., Wynne N., Chandrasegaran K., Wolff G., Vinauger C. Modulation of mosquitoes' responses to visual threats – **Entomology**, Portland, OR — ***Oral presentation***

Dessart M., Luff S., Smith L., Sunman H., Vinauger C. The plasticity of mosquito circadian clock – **Animal Behaviour Live** — ***Online poster presentation***

Dessart M., Luff S., Smith L., Sunman H., Vinauger C. How to jetlag a mosquito – **CeZAP Infectious Disease Research Symposium**, Blacksburg, VA — ***Poster presentation***

2024 **Dessart M.**, Lazzari C., Guerrieri F. The cognition of mosquito larvae as a biological indicator of environmental quality – **French Society of Neuroethology**, Albi, France — ***Oral presentation***

Dessart M., Lazzari C., Guerrieri F. La cognition des larves de moustique comme indicateur biologique de la qualité des écosystèmes aquatique - **Insect Biology Research Institute**, Tours, France — ***PhD Defense***

2023 **Dessart M.**, Lazzari C., Guerrieri F. Cognitive abilities in mosquito larvae as indicators of environmental quality – **Behaviour**, Bielefeld, Germany — ***Oral presentation***

Dessart M., Lazzari C., Guerrieri F. Chemical pollution affects the behaviour of mosquito larvae – **DiPEE Biodiversité et territoires**, Chizé, France — ***Oral presentation***

Dessart M., Lazzari C., Guerrieri F. Chemical pollution affects the behaviour of mosquito larvae – **French Society of Animal Behavior**, Tours, France — ***Poster presentation***

2022 **Dessart M.**, Lazzari C., Guerrieri F. Effect of two herbicides and one medical drug on locomotor activity and learning in *Aedes aegypti* – **Colloque Biotechnocentre**, Centre Val de Loire, France — ***Oral presentation***

Dessart M., Pineirúa M., Lazzari C., Guerrieri F. A tracking system to assess activity and learning in mosquito larvae – **Ecology and Behaviour**, Strasbourg, France — ***Poster presentation***

2021 **Dessart M.**, Robineau T., Lazzari C., Guerrieri F. Visual learning abilities in mosquito larvae are not impaired neither by light deprivation nor by turbidity – **European Network Insect Science**, Florence, Italy — ***Oral presentation***

Dessart M., Aguiar J., Tabacchi E., Guillerme S., Giurfa M. Invasive plant species through the bee eye: an analysis of flower coloration in the French Pyrenees – **Hubert Curien pluridisciplinary Institute**, Strasbourg, France — ***Invited talk***

Dessart M., Lazzari C., Guerrieri F. The influence of physico-chemical characteristics of aquatic ecosystems on prey–predator interactions. A behavioural-neuroanatomical approach – **Insect Biology Research Institute**, Tours, France — ***Invited talk***

Dessart M., Aguiar J., Tabacchi E., Guillerme S., Giurfa M. Invasive plant species through the bee eye: an analysis of flower coloration in the French Pyrenees **French Society of Animal Behavior**, Marseille, France — ***Poster presentation***

2020 **Dessart M.**, Bastos A., Taylor A. Effects of the warble call on optimism in Kea (*Nestor notabilis*) – **Research Center on Animal Cognition**, Toulouse, France — ***Invited talk***

Academic and Professional Honors

2026 Travel grant to Internation Congress of Neuroethology 2026, Vancouver, Canada - Office of Research and Innovation, Virginia Tech

2025	Travel grant to Entomology 2025, Portland, OR - Office of Research and Innovation, Virginia Tech
2024	Travel grant – French Society of Animal Behavior (SFECA), Albi, France
	Ph.D. obtained with High Honors, University of Tours, France
2023	Travel grant – Behaviour Conference, Bielefeld, Germany - University of Tours
2022	Regional finalist of My thesis in 180 seconds program, University of Tours, France
2021	3-years doctoral Fellowship, Region Centre Val de Loire, University of Tours, France
2019	Master's degree obtained with High Honors, University of Strasbourg, France
	Travel grant – Region Alsace GrandEst, Strasbourg, France
2017	Scholarship – 9-month training Udacity & Google online formation
	Engineering degree obtained with High Honors, Institut Catholique des Arts et Métiers, Toulouse, France

Academic & Professional Service

2026 - Present	Member of the communication team at *Animal Behavior Live*
2025 - Present	Review of manuscripts for *Scientific Reports*, *Ecology and Evolution*, *Journal of Experimental Biology*, *PLOS Biology*
2025 - 2026	Board member of the *Postdoc & Research Faculty Committee* in the Department of Biochemistry, Virginia Tech
	Board member of the *Virginia Tech Postdoctoral Association*, Virginia Tech
2020 - 2021	Contributed to the writing of a successfully funded 3-years PhD project (IPPY)
2013 – 2016	Communication Officer, Student Office & Secretary, Student Music Association *Institut Catholique des Arts et Métiers* — Toulouse, France

Memberships

2023 - Present	Member of the *French Society of Neuroethology*
	Member of the *International Society of Neuroethology*
2021 - Present	Member of the *French Society of Animal Behavior*
2025 - 2026	Member of the *Entomological Society of America*
2023 - 2024	Member of the *German Society of Ethology (Ethologische Gesellschaft)*
2021 - 2024	Member of the *French Union for the Study of Social Insects (UIEIS)*

Training & Volunteering

02/26 - 03/26	Training: Crucial Conversations Book Club - Virginia Tech, VA
04/2023	Training: *Process Communication Model* - CNRS, University of Tours, France
01/2023	Training: One-week training on statistical modeling - IRD, Montpellier, France
03/18 - 04/18	Volunteering: Assisted a vet and conducted interviews with farming communities. *Melindika Solidarity Association* - Kafue National Park, Zambia
01/18 - 02/18	Volunteering: assisting a PhD student with a protocol on cats to study their biological rhythms. *AVA Sanctuary* - Normandie, France
10/16 - 12/16	Volunteering: Assisted in the realisation of cognitive enrichment items for primates. *Auckland Zoo* - Auckland, New Zealand
2014 – 2017	Volunteering: Activity leader in France, Morocco, and Madagascar. *Terres en Mêlées Solidarity Association* — Toulouse, France